

Milestone Delivery #4
CSE-2102 Group Project - Spring 2025
Design Artifacts
Deadline: 11:59pm, March 27

Pre-Requisite

- Feature list from Milestone #2
- Figma designs from Milestone #3

Project Board

- Create cards and assign them to teammates for all work relating to this milestone delivery.
- Keep your project board up-to-date

Design Artifacts

- Your team will create various design diagrams to document your design. We call these "artifacts" because they will remain as important pieces of information that continue to have value long after your system is implemented.
- Make them as complete as possible based on what you know about your project so far. You will continue to update them throughout the semester.
- Your scenarios, user stories, feature list, and Figma UX prototype will inform the detail you put into these diagrams. These diagrams should naturally flow from the work you have already done.
- Use DrawIO for all diagrams. Name the files using this convention, where NN is two-digit number (01, 02, 03, ...) and xxxxxxxx is a meaningful name that represents what the diagram is depicting:
 - seq-NN-xxxxxxx
 - usecase-farm-market
 - er-farm-market
- Use Case Diagram (1 diagram)
 - Identity the different types of users in your system
 - Create ovals for every major action those users will take in your system
 - Associate the users to the correct actions in your diagram
- Sequence Diagram (1 sequence diagram for every distinct flow)
 - Identify the major flows of your system. For instance, one flow might be the steps through the system when someone places an order. Another flow might be the login process.

- Draw a sequence for every major and unique flow through the system. If you find that two flows have the identical steps and path, create a single sequence and name it something more generic that reflects that fact.
- Name each diagram something that reflects what flow it is depicting.
- You should have at least 4 sequence diagrams.
- ER Diagram (1 diagram)
 - Create a single ER Diagram that reflects your data model for the project. Use the following bullets as steps in creating the required detail for the diagram:
 - Identity the types of data you will need in your database.
 - Categorize your data into groupings or "tables".
 - Identify what data type is most appropriate for each table.
 - Identify the primary key for each table. Note: you may need to add a simple integer field to a table to make the rows unique.
 - Identify the foreign keys that will link the tables together. Determine the cardinality of those relationships. (E.g. is it one-to-one, one-to-many, or many-to-many?) Show the linkage lines and the cardinality on your diagram.

Git

- Create a feature branch
- Add the source DrawIO files (not a PDF or screenshot!) to the "docs" folder. You should have a separate file for every sequence diagram, one for the use case diagram, and one for the ER Diagram.
- Push the feature branch. Open a PR for merging to develop. Open a PR for merging to main.

Grading

- One team member will submit a PDF version of the Sequence Diagram to HuskyCT. (One submission per team.) You will be graded by what is in your git repo "docs" folder. You are simply uploading to HuskyCT so that TAs can leverage HuskyCT for grading purposes.
- Make sure you have all your source files in the "docs" folder on git.
- You may submit multiple time prior to the deadline. It is strongly recommended you do so in case you run into any last minute technical or team issues.
- You will meet with a TA during your lab. Your team may not yet be complete with all the milestone tasks, but your TA will be looking at:
 - Current status of your project board
 - Current status of your artifacts
 - Did your artifacts flow from your previous milestones
 - Team collaboration and communication.

- Reminder: you must use DrawIO to draw your diagrams. The use of AI image creation tools is not allowed.